

EXPERIMENT : 12Preprocessor and Directives in C

- 1) Write a program to define some constant variable in processor

```
#include <stdio.h>
```

```
#define PI 3.14159
```

```
#define MAX_SIZE 100
```

```
#define COUNTRY "India"
```

```
#define YEAR 2025
```

```
int main ( ) {
```

```
    printf ("Value of PI : %.5f\n", PI);  
    printf ("Maximum Size : %d\n", MAX_SIZE);  
    printf ("Country Name : %s\n", COUNTRY);  
    printf ("Current Year : %d\n", YEAR);
```

```
    return 0;
```

```
}
```

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Output :- Value of PI : 3.14159

Maximum Size : 100

Country Name : India

Country Year : 2025

Teacher's Signature: _____

main.c



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Output

Clear

```
1 #include <stdio.h>
2
3 #define PI 3.14159
4 #define MAX_SIZE 100
5 #define COUNTRY "India"
6 #define YEAR 2025
7
8 int main() {
9
10     printf("Value of PI      : %.5f\n", PI);
11     printf("Maximum Size    : %d\n", MAX_SIZE);
12     printf("Country Name    : %s\n", COUNTRY);
13     printf("Current Year    : %d\n", YEAR);
14
15     return 0;
16 }
17 |
```

```
Value of PI      : 3.14159
Maximum Size    : 100
Country Name    : India
Current Year    : 2025
```

```
=== Code Execution Successful ===
```



2) Write a program to define a function in directives.

```
#include <stdio.h>
```

```
#define ADD(a,b) (a)+(b)
```

```
#define MULTIPLY(x,y) (x)*(y)
```

```
#define GREET() printf("Hello! This is a  
function defined in directives.\n")
```

```
int main () {
```

```
    int x = 10, y = 20;
```

```
    GREET();
```

```
    printf("Addition of %d and %d = %d\n",  
           x, y, ADD(x,y));
```

```
    printf("Multiplication of %d and %d = %d\n",  
           x, y, MULTIPLY(x,y));
```

```
    return 0;
```

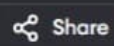
```
}
```

Output :- Hello! This is a function defined in directives

Addition of 10 and 20 = 30

Multiplication of 10 and 20 = 200

main.c



Run

Output

Clear

```
1 #include <stdio.h>
2
3 #define ADD(a, b) ((a) + (b))
4 #define MULTIPLY(x, y) ((x) * (y))
5 #define GREET() printf("Hello! This is a function defined in directives.\n")
6
7 int main() {
8
9     int x = 10, y = 20;
10
11     GREET();
12     printf("Addition of %d and %d = %d\n", x, y, ADD(x, y));
13     printf("Multiplication of %d and %d = %d\n", x, y, MULTIPLY(x, y));
14
15     return 0;
16 }
17
```

Hello! This is a function defined in directives.
Addition of 10 and 20 = 30
Multiplication of 10 and 20 = 200

=== Code Execution Successful ===